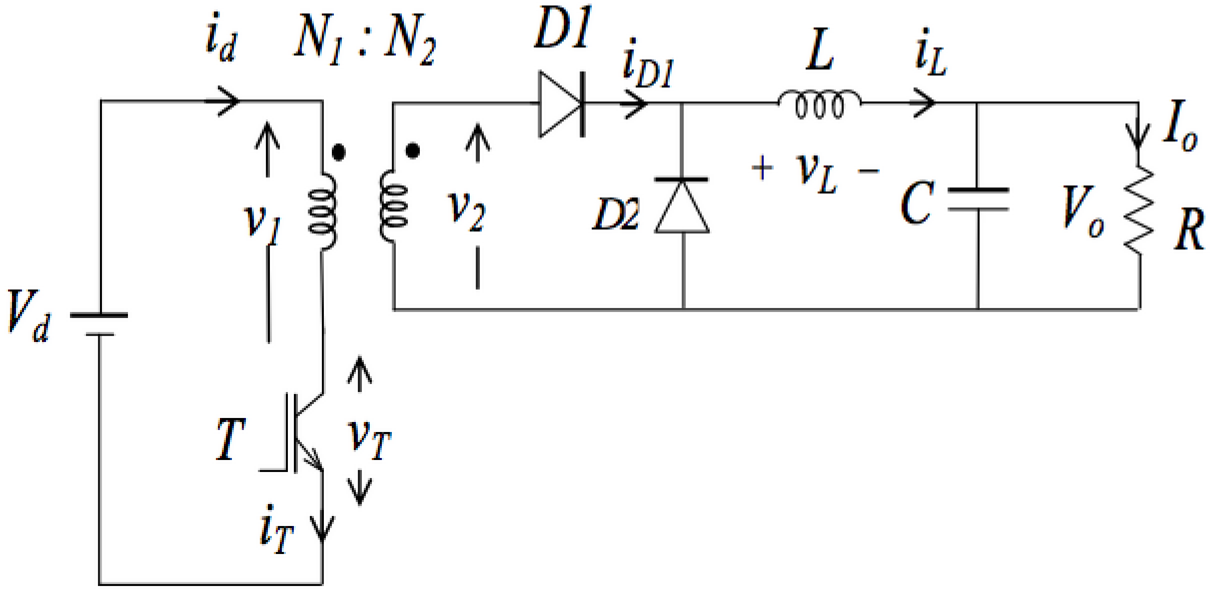
Forward converter

DC-DC CONVERTER

Introduction

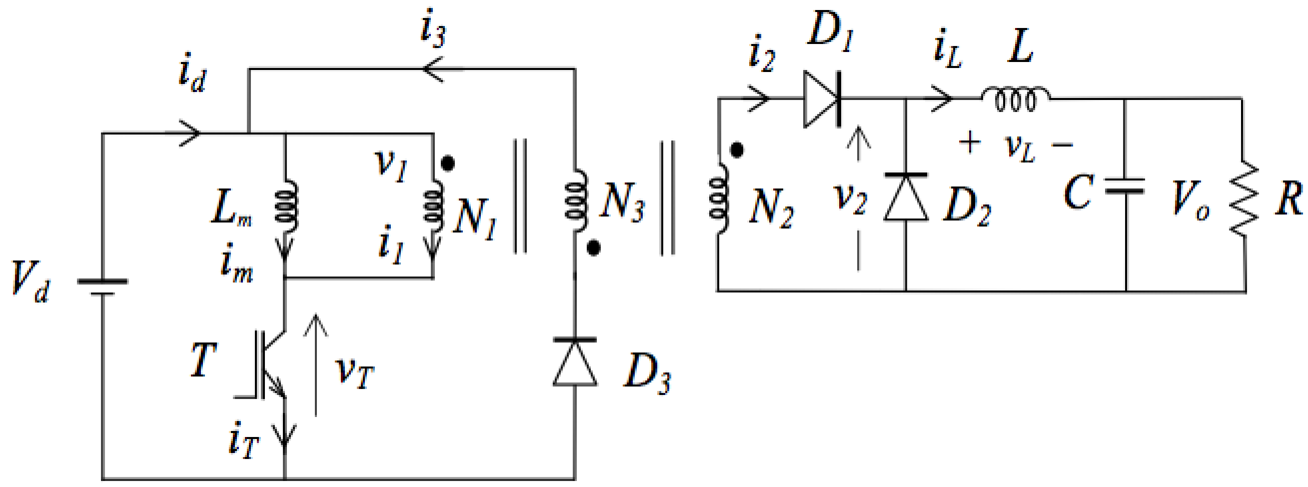
The forward converter is really just a transformer isolated buck converter. Like the flyback topology, the forward converter is best suited for lower power applications.



While efficiency is comparable to the flyback, it does have the disadvantage of having an extra inductor on the output and is not well suited for high voltage outputs.

The forward converter does have the advantage over the flyback converter when high output currents are required. Since the output current is non-pulsating, it is well suited for applications where the current is in excess of 15A.

Forward converter with demagnetizing winding and energy return diode



The introduction of a third winding and energy return diode improves the forward converter in two respects. Firstly it provides a pathway for the energy stored in Lm to return and secondly it also helps to reset the transformer flux quickly.

The third winding is only necessary when you're using a real transformer model. The third winding + a third diode are used to demagnetize the core. In the case when you're using an ideal transformer model there is no such need and the converter works just like a buck converter (but with the transformer relation)

What´s the difference between flyback and forward converter?

The difference between flyback vs. forward converters lies in the inductive energy storage.

In the flyback converter, the energy storage is the transformer itself, which is why a transformer with an air gap is needed.

The forward converter uses a transformer without an air gap, so an additional storage choke is needed. The forward converter is therefore somewhat more complex in design, but also achieves a higher efficiency.

Design:

Vi=170 v
Vo=5 v
F=300 kH
iL=5 A
Power=100 w
R=400 ohm

Step 1: determine the duty ratio value:

$$\frac{N1}{N2} = \frac{Vi}{Vo} D$$

The turns ratio is 12 so the duty will be 0.353

Step 2: determine the L value:

Assume 40% variation in inductor current so:

$$\Delta il = 0.4 * 5 = 2\ A$$

$$L = \frac{(1 - D) * vo}{\Delta il * f}$$

Step three: determine the C value:

Assume 1% ripple in output voltage so:

$$\Delta vo = 0.01 * 5 = 0.05\ v$$

$$\Delta vo = \Delta ic * rc = 0.05$$

$$2 * rc = 0.05$$

$$rc = 25\ m\ ohm$$

$$C = \frac{10^{-5}}{0.025} = 400\ u\ ohm$$

```
vi=170;
vo=5;
f=300000;
il=5;
R=400
power=100
variation_il=0.4;
turn_ratio=12;
d=(turn_ratio*vo)/vi

delta_il=variation_il*il
l=((1-d)*vo)/(delta_il* f)
delta_vo=0.01*vo
rc=delta_vo/delta_il
c=(10^-5)/rc
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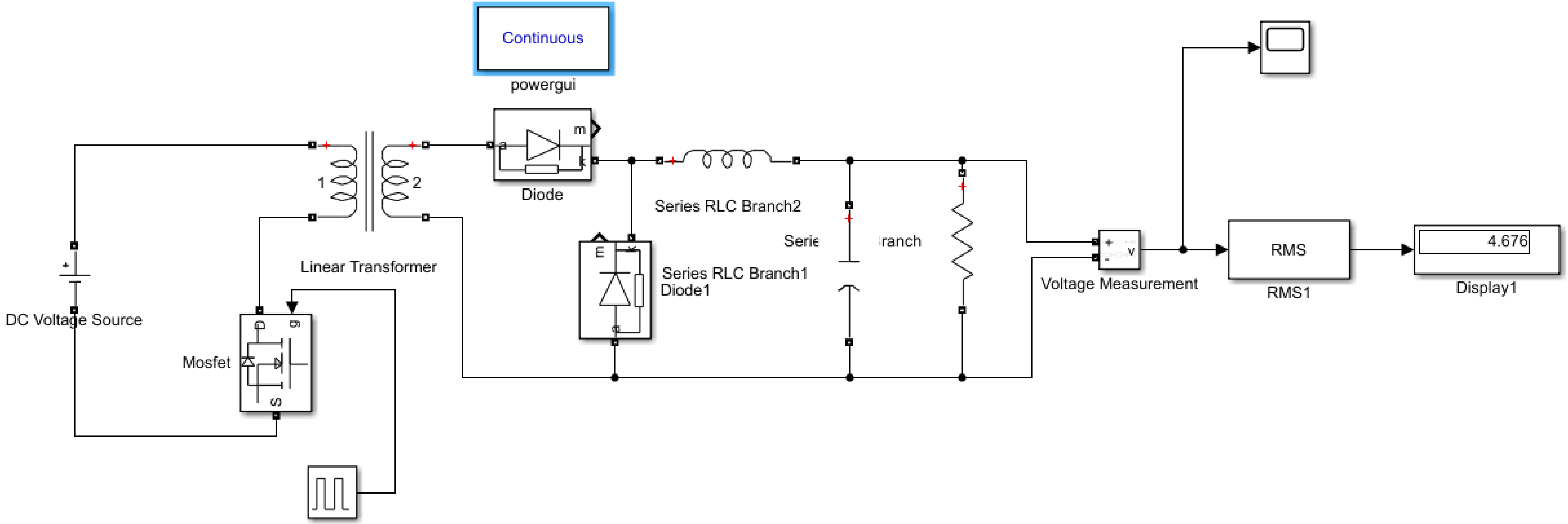
$$\frac{N1 * vo}{N2 * vi} = D$$


Continuous
powergui
Diode
m
Linear Transformer
1
2
DC Voltage Source
Mosfet
Series RLC Branch2
Series RLC Branch1
Diode1
Voltage Measurement
RMS
RMS1
4.676
Display1